

ANEESH GROVER

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Experience

AcademicAlly

July 2024 – August 2025

Founding Developer – ACM Thapar Society

React, Python, Pandas, Openpyxl, BeautifulSoup

- **Served 10,000+** students by developing a responsive timetable portal for fast, reliable class schedule lookup.
- Built ETL pipelines to transform and validate **50,000+** Excel rows into optimized JSON datasets, implementing custom error handling that reduced parsing failures by **95%**.
- Supported deployment and maintenance of the **live portal**, coordinating with team members.

Projects

FocusBoard - Whiteboard Task Capture System

Kotlin, FastAPI, CRAFT, TrOCR, Phi-3 Mini, TFLite

- Developed the Android app in Kotlin with ML Kit document scanning and an experimental **on-device YOLOv26n/TFLite** whiteboard detector, chosen for mobile size constraints.
- Engineered a **3-stage OCR pipeline** using CRAFT for line segmentation, TrOCR for transcription, and Phi-3 Mini for structured calendar JSON generation from handwritten input.
- Designed per-device request queues to serialize uploads, preventing concurrent TrOCR worker spawning and **eliminating cross-device race conditions** under simultaneous load.
- Returned structured JSON to the app for user approval before committing tasks to the shared E Ink display.

Campus-Compute - Distributed Compute Framework

Python, FastAPI, WebSockets, Docker

- Architected a LAN-distributed compute system with a centralized coordinator and worker nodes, enabling automatic discovery and reconnect handling.
- Designed a capability-aware scheduler using CPU, RAM, and GPU metadata to optimize task placement across nodes.
- Implemented a real-time WebSocket dashboard for cluster telemetry, job orchestration, and containerized workload execution.

folio.mcp - Personal MCP Server

TypeScript, Cloudflare Workers, Hono, MCP SDK

- Deployed a remote MCP server on Cloudflare Workers exposing GitHub activity, portfolio data, and an LLM-invocable contact tool.
- Added a KV layer for GitHub response caching with TTLs and IP-based rate limiting on the contact tool to prevent abuse.
- Shipped a companion frontend at mcp.aneeshgrover.me  with dynamic tool discovery; automated deploys via GitHub Actions.

Orbital - Cloud IDE Control Plane

Go, Docker, JWT, REST APIs

- Architected a control plane in Go that provisions isolated VS Code workspaces on demand via a Dockerized provisioner.
- Implemented a reconciler loop to track workspace state and recover from provisioning failures without manual intervention.
- Secured the API surface with JWT-based authentication and a layered internal package structure separating auth, storage, and orchestration concerns.

Skills

Languages: Python, Go, Java, C++, JavaScript, TypeScript, SQL

Interfaces for Agents: Model Context Protocol (MCP), LLM tool integration, Cloudflare Workers, Hono, REST APIs

Infrastructure for Developers: distributed systems, WebSockets, JWT authentication, system design, Docker, CI/CD, Git

Systems that Learn: PyTorch, TensorFlow, transformers, RAG, vector databases, OpenCV, CUDA

Data & Platforms: Supabase, PostgreSQL, Pandas, asynchronous processing

Education

University of Toledo

Toledo, OH

Bachelor of Science in Computer Science and Engineering

GPA: 3.5/4.0

Leadership & Involvement

Association for Computing Machinery (ACM)

Nov 2023 – Present

Executive Board Member & Machine Learning Bootcamp Lead

- Organized Hacklipse 2024, ACM Thapar's flagship hackathon, coordinating operations for **300+** participants.
- Mentored **100+** students through ACM's Machine Learning Bootcamp by designing curriculum and leading weekly technical sessions.

Achievements

- Secured **runner-up** in **Open Innovation Category** at **NVIDIA's SatHack**, competing against **1500+** participants.
- Awarded Best Use Case for AcademicAlly at ACM Projectathon 2024.